Ho Chi Minh University of Science
Department of Information Technology

Data Visualization - 23KHDL02

LAB 01 REPORT

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1. Introduction

The goal of this lab is to systematically collect, analyze, and visualize product data from e-commerce platforms. Currently, e-commerce data is fragmented across individual product pages and lacks systematic organization. By aggregating this data, we can gain a comprehensive overview of the online retail market, understand consumer behavior, and draw practical business conclusions.

This lab has the following specific objectives:

- Build a well-structured, sufficiently large, and analytically meaningful e-commerce dataset.
- Practice real-world data collection techniques, specifically web crawling, rather than using pre-existing datasets.
- Apply data visualization techniques to explore and demonstrate the relationships between different market variables.
- Define a general analytical problem for the team, with each member establishing at least two specific, measurable (SMART) objectives to address it.

2. Assignment

Name	Task	Evaluation	Overall project contribution
Nguyễn Hoàng Minh Tâm	- Report Structuring - Objective 8, 9, 10	100% 100%	20%
Nguyễn Đức Khương Lam	- Cleanup data - Implementation of Machine Learning - Objective 4, 11	100% 100% 100%	20%
Trần Nguyễn Song Chi	- Setting up the database and planning the schema - Objective 1, 5	100% 100%	20%
Nguyễn Hiệp Thắng	- Crawling data - Objective 2, 3	100% 100%	20%
Trần Dương Hải Thượng	- Dashboard designing - Objective 6, 7	100% 100%	20%

3. Data and objective Preparation

3.1 Data Crawling

3.1.1 URL harvesting

Goal: Collect relevant links from target websites

Technology: Selenium and **BeautifulSoup**. **Selenium** simulates human interaction to bypass dynamic rendering and lazy loading (e.g., "Load More" buttons) that standard HTTP requests cannot access.

Approach:

1. Dismiss pop-ups and advertisements.
2. Navigate to the target product category.
3. Extract URLs matching specific keywords (e.g., "laptop") and filter out exclusions (e.g.,
"balo", "chuot").
4. Click "Load More", scroll to trigger lazy loading, and append new links.
5. Repeat until the end of the page is reached.

Implementation: A single, universal “Crawler” class handles all websites. Site-specific elements are injected via a configuration dictionary rather than hardcoded logic.
Example Configuration Profile:

```
{
   "source_name": "FPT Shop",
   "base_url": "https://fptshop.com.vn/",
   "start_url": "https://fptshop.com.vn/may-tinh-xach-tay",
   "link_keywords": ["laptop", "macbook"],
   "exclude_keywords": ["balo", "chuot", "cap", "phu-kien"],
   "load_more_xpath": "//button[contains(., ’Xem them’)]",
   "popup_selectors": [".close-popup", "#onesignal-cancel"]
}
```

Benefit: High scalability. Adding a new website requires only a new configuration block, avoiding redundant code.

Output: A structured JSON array containing basic product metadata (source, initial name, and target URL) ready for the extraction phase.

Example Output:


```
[
  {
    "source": "An Phat PC",
    "name": "Laptop Hp Elitebook 6 G1I 13 Bq9M7Pt",
    "url": "https://www.anphatpc.com.vn/laptop-hp-elitebook-6-g1i-13-bq9m7pt.html"
  },
  ...
]
```


3.1.2 Data Extraction

Goal: Extract detailed metadata, specifications, pricing, and social metrics from the collected links.

Technology: requests, **BeautifulSoup**, and **concurrent.futures.requests** replaces **Selenium** for faster static HTML fetching. Multithreading enables concurrent scraping. Custom User-Agent headers and randomized delays prevent IP blocking.

Approach: Send HTTP requests to collected URLs, parse the source code (HTML or hidden JSON objects), extract specific data components, and export the standardized data to JSON.

Implementation:

- **Base Class (BaseDeepScraper):** Handles common utilities (HTTP requests, error handling, ID generation).
- **Derived Classes (e.g., TGDDScraper):** Inherit from the base class and execute site-specific parsing logic.

Benefit: Separation of concerns. Layout changes on one website only require updating its specific derived class, leaving the core engine and other scrapers unaffected.

Output: A comprehensive JSON array containing the raw, extracted hardware specifications, pricing data, and social metrics for each product.

Example Output:


```
[
  {
    "id": "anphat_laptop-asus-expertbook-p1503cva-c5h08-50w",
    "metadata": {
      "source": "An Phat PC",
      "crawl_date": "2026-03-21T18:38:58",
      "url": "https://www.anphatpc.com.vn/laptop-asus-expertbook-p1503cva-c5h08-50w.html"
    },
    "product": {
      "name": "Laptop ASUS ExpertBook P1503CVA-C5H08-50W",
      "is_available": true
    },
    "pricing": {
      "current_price": 18690000
    },
    "specs_raw": {
      "CPU": "Intel Core 5 Processor 210H 2.2 GHz",
      "VGA": "Intel Graphics",
      "Display": "15.6 inch FHD (1920 x 1080) 16:9",
      "Ram": "8GB",
      "Ổ cứng": "512GB M.2 2280 NVMe PCIe 4.0 SSD",
      "Pin": "3-cell 63WHrs",
      "Weight": "1.68 kg",
      "Color": "Misty Grey",
      "OS": "Windows 11 Home"
    },
    "social": {
      "avg_rating": 0.0,
      "review_count": 0,
      "satisfied_info": "615 Lượt xem"
    }
  },
  ...
]
```

3.2 Data Transformation & Integration

3.2.1 Standardization and Extraction

Goal: Convert messy, unstructured text into standardized categories and numeric values.

Approach:

- **Text Normalization:** Accents are removed and text is lowercased. A mapping dictionary

is used to unify different retailer terms into a single standard key (e.g., mapping "vi xử lý", "chip", and "bo xu ly"to "cpu").

- **Regex Extraction:** Regular Expressions (Regex) are applied to isolate exact numeric

values from descriptive strings. For example, it extracts 16 and DDR5 from the string "16GB LPDDR5 7500MHz".

- **Categorical Binning:** Continuous or highly varied data is grouped into standard buckets

to simplify filtering. For example, exact prices are grouped into segments like "Budget"or "Premium", and screen sizes are grouped into standard ranges (e.g., 15.6 16.0").

3.2.2 Missing Data Imputation

Goal: Fill in missing data points (such as weight or battery capacity) without deleting the entire product record.

Approach: A custom Data Imputer class groups laptops by shared contextual features (e.g., Brand, Product Series, and Price Segment). If a specific laptop is missing its weight, the imputer calculates the statistical mode (the most common value) of similar laptops within that group and uses it to fill the blank.

Benefit: This cross-retailer checking recovers critical data and prevents significant data loss, ensuring a more accurate final dataset.

3.2.3 Relational Modeling

Goal: Eliminate data redundancy and format the dataset for Power BI and Supabase.

Approach: The cleaned, flat dataset is split into three separate tables linked by unique IDs (Foreign Keys).

- **Dim_Source (Dimension):** Contains unique retailer identifiers (e.g., "S01": "The Gioi Di Dong").

- **Dim_Product (Dimension):** Contains unique hardware configurations. If five retailers sell the exact same laptop, its specifications are only stored once here.
- **Fact_Source_Product (Fact Table):** The bridge table connecting the source and the product. It stores the dynamic metrics: current price, average rating, and review counts.

Benefit: This structure maintains data integrity, reduces file size, and allows dashboard filters to work efficiently across multiple retailers.

3.3 Determining Objectives

The determination of the objectives is based on the role of users and the problem statement. In this project, we are focusing on designing the dashboard suitable for Tech-Retail Strategy Consultant, which allows them to optimize pricing strategies and product inventory across six major retail chains in Vietnam to capture market share. Therefore, we have a list of objectives divided into 6 main themes:

Theme 1: Price and Competitive Landscape

Objective 1: Compare product category concentration among competitors

- Selected Metrics: Laptop count and Category (Gaming/Office/Creator).
- Metric Meaning: The number of laptop models allocated by use-case at each retail chain.
- Rationale: Accurately identifies the primary competitor in specific market segments.
- Chart Choice: Clustered Column Chart.

Objective 2: Evaluate market share by price segment

- Selected Metrics: Model count by price brackets (Under 15m, 15-25m, 25-45m, Over 45m VND).
- Metric Meaning: The proportion of inventory categorized by price level.
- Rationale: Reveals whether a competitor focuses on budget-friendly or premium customer segments.
- Chart Choice: 100% Stacked Column Chart.

Objective 3: Assess overall pricing strategy

- Selected Metrics: Current price and Source (Retail Chain).
- Metric Meaning: The distribution of prices (minimum, maximum, and median) for all products in a store.

- Rationale: Instantly identifies pricing positioning (budget vs. premium) and price spread.
- Chart Choice: Boxplot.

Theme 2: Technology Transition ("AI Laptop")

Objective 4: Price the value of AI features (AI Premium)

- Selected Metrics: Current price and AI Label (is_ai_laptop: True/False).
- Metric Meaning: The average price difference between machines with an NPU (AI) and standard machines.
- Rationale: Evaluates if the price markup for AI features is reasonable.
- Chart Choice: Grouped Bar Chart.

Theme 3: Hardware Trends

Objective 5: Track the shift toward 32GB RAM

- Selected Metrics: RAM capacity (16GB vs. 32GB) and Price Segment.
- Metric Meaning: The percentage of laptops using 32GB RAM within each price bracket.
- Rationale: Informs inventory decisions regarding the new RAM standard.
- Chart Choice: 100% Stacked Bar Chart.

Theme 4: Product and Brand Strategy

Objective 6: Identify brand strengths

- Selected Metrics: Model count, Brand, and Category.
- Metric Meaning: The distribution of a brand's product lines (e.g., % of Gaming vs. Office).
- Rationale: Advises on brand selection for specific market niches.
- Chart Choice: 100% Stacked Column Chart.

Objective 7: Categorize brands by Price and Configuration

- Selected Metrics: Median price and Average RAM per brand.
- Metric Meaning: Representative cost and performance per manufacturer.
- Rationale: Clusters brands into "Budget," "Mid-range," and "Premium" for inventory alignment.
- Chart Choice: Scatter Plot.

Theme 5: Consumer Demand

Objective 8: Identify the most preferred brands by demand

- Selected Metrics: Satisfied customer info, Brand, and Category.
- Metric Meaning: Total engagement and satisfaction levels.
- Rationale: Acts as a "sales proxy" for stocking decisions.
- Chart Choice: Bar Chart.

Objective 9: Validate actual market demand for RAM upgrades

- Selected Metrics: Satisfied customer info and RAM capacity.
- Metric Meaning: Customer interest and willingness to pay for high-RAM machines.
- Rationale: Verifies if inventory trends match real consumer buying behavior.
- Chart Choice: Donut Chart.

Objective 10: Evaluate satisfaction by price segment

- Selected Metrics: Average rating, Satisfied customer info, and Price Segment.
- Metric Meaning: Expectations met relative to cost.
- Rationale: Identifies market gaps in low-satisfaction segments.
- Chart Choice: Bubble Chart.

Theme 6: Predictive Pricing & Strategy (Machine Learning)

Objective 11: Identify "Best Value" laptops at Phong Vu using Random Forest Regression

Goal & Purpose: Train a Random Forest Regression model to calculate the "Fair Market Price" of laptops based on hardware specifications (RAM, CPU, GPU, Weight, and AI features). The model acts as a tool to automatically detect which models at Phong Vu are priced significantly lower than their hardware's predicted market value.

Actionable Examples:

- Scenario A: Store Price > Predicted Price. Action: Apply discount.
- Scenario B: Store Price < Predicted Price. Action: Tag as "Best Deal".

Assumptions & Limitations: Requires stable markets; needs monthly retraining with fresh data.

- Selected Metrics: Actual Store Price, Predicted Market Price, and "Bargain Value" (Predicted Price minus Actual Price).
- Metric Meaning: The "Bargain Value" represents the financial saving a customer receives relative to the hardware's estimated market worth.

- Rationale: Identify undervalued inventory, providing Phong Vu with data-backed promotional opportunities.
- Chart Choice: Horizontal Bar Chart (ranking the top 10 laptop models at Phong Vu with the highest "Bargain Value").

4. Analysis

4.1. Dataset Description

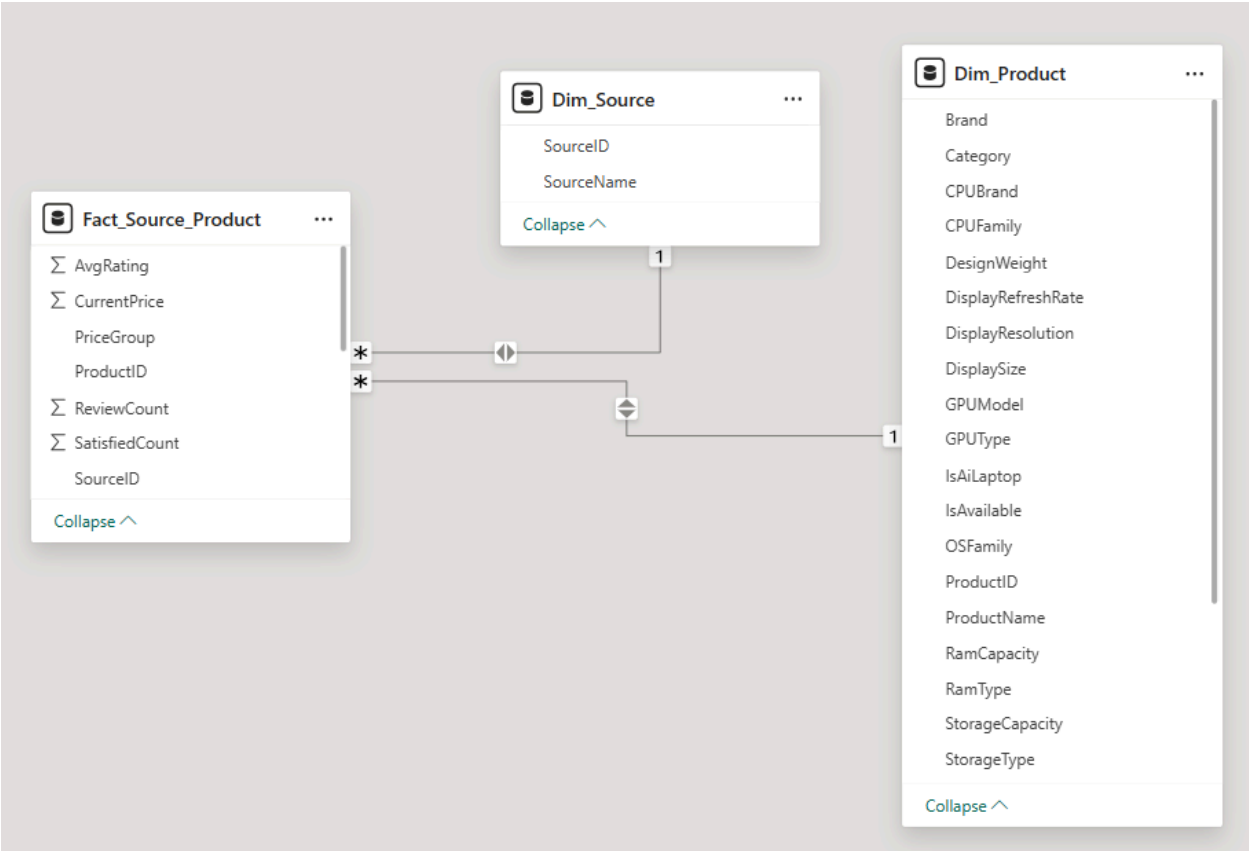

Figure 1: Relational model

Dim_Product: This table stores all the details and technical specifications of the laptops

- ProductID: A unique identifier for each laptop model
- ProductName: The full name of the laptop as listed by retailers
- Brand: The manufacturer of the laptop
- Category: The intended use-case or target audience
- CPUBrand: The manufacturer of the processor
- CPUFamily: The specific line of the processor
- GPUType: Whether the graphics card is integrated or dedicated
- GPUModel: The specific category or model of the graphics card
- RamCapacity: The amount of RAM in the laptop, measured in gigabytes (GB)

- RamType: The generation of the memory
- StorageCapacity: The size of the storage drive, measured in gigabytes (GB).
- StorageType: The kind of storage drive used
- DisplaySize: The physical diagonal size of the screen in inches
- DisplayResolution: The category of screen resolution
- DisplayRefreshRate: The screen's refresh rate in Hertz
- OSFamily: The operating system pre-installed on the machine
- DesignWeight: The weight of the laptop
- IsAiLaptop: Whether the laptop has AI features or not
- IsAvailable: Whether the laptop is currently active or not

Dim_Source: This table stores information about the various retail platforms selling these laptops

- SourceID: A unique identifier for the retailer
- SourceName: The name of the retail store or e-commerce platform

Fact_Source_Product: This table stores metrics like prices and reviews of a specific laptop at a specific retailer

- SourceID: A unique identifier for each laptop model (links back to the retailer in Dim_Source)
- ProductID: A unique identifier for the retailer (links back to the specific laptop in Dim_Product)
- AvgRating: The average customer rating out of 5 stars for that laptop on the retailer's site
- ReviewCount: The total number of user reviews left for the laptop on the retailer's site
- SatisfiedCount: The number of positive ratings
- CurrentPrice: The current selling price of the laptop at that retailer in Vietnamese Dong (VND)

4.2. Analysis of Data

Objective 1: Comparison of the types of laptop that each competitor sells

We chose this objective because it helps determine the best competitor in each category of laptops.

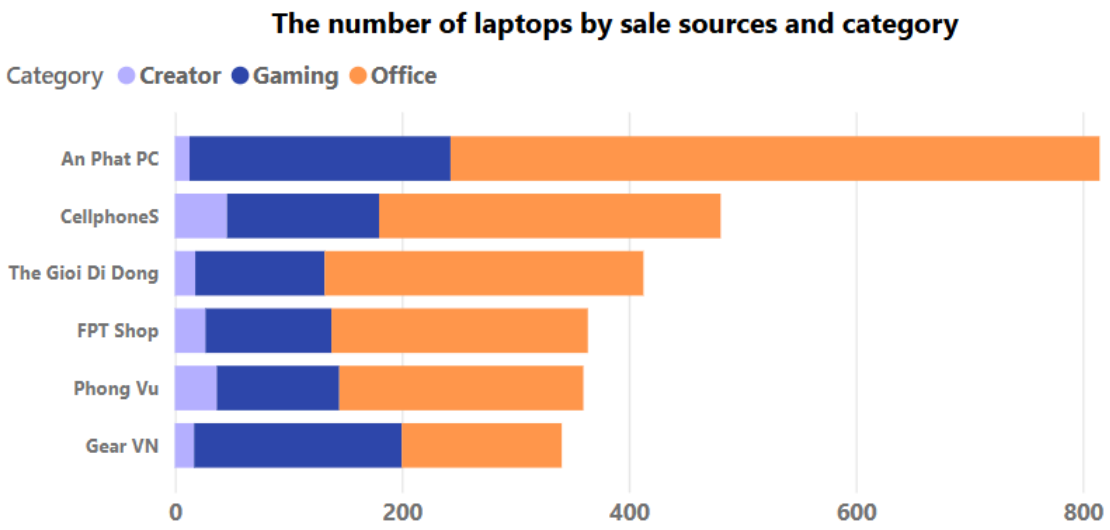

Figure 2: The number of laptops available, grouped by source and category

Regarding the total number of laptops sold at each retailer, An Phat PC was the most dominant competitor with the largest total of over 800 laptops. This was followed by that of CellphoneS, TheGioiDiDong at around 480 and 410 laptops respectively. Finally FPT Shop, Phong Vu and Gear VN had around 350-375 laptops each.

An interesting takeaway from the chart is that the majority of laptops sold at all retailers belonged to the Office category, which was followed by the Gaming and Creator categories. The only exception is in Gear VN, where Gaming laptops held the largest share of the type of laptops sold. Regarding Office and Gaming laptops, An Phat PC had the highest number of these laptops, especially Office laptops, since its figure was higher than that of the runner up CellphoneS by a margin of 200 laptops. Meanwhile, CellphoneS had the highest number of Creator laptops.

Objective 2: Market analysis on the price range

The aim of this objective is to find out which type of customers (poor, middle or upper class) each retailer was catering to.

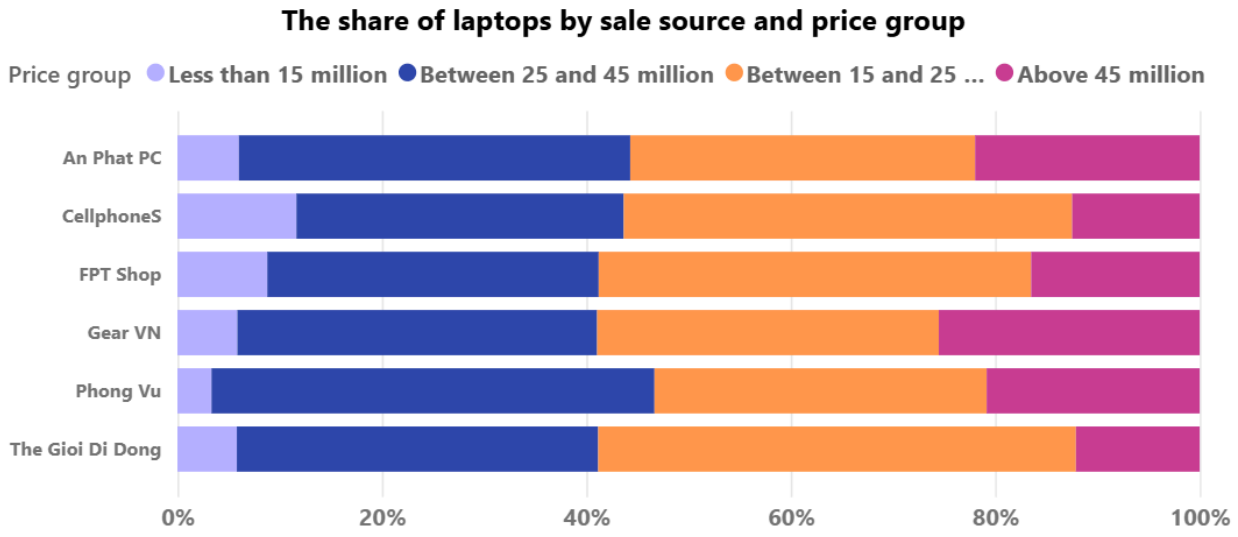

Figure 3: The distribution of laptops based on their retailers and price groups

All retailers shared similarities in the distribution of both the share of their products below 15 million VND and that above 45 million VND (these figures remained the 2 lowest price ranges across the board).

For An Phat PC, Gear VN, Phong Vu, the share of products in the price range of 25-45 million VND was higher than that of the price range 15-25 million VND, while the opporetailer was true for the remaining retailers.

Regarding the products in the price range of 25-45 million VND, Phong Vu had the highest figure at 43,3%, which was followed by AN Phat PC, and The Gioi Di Dong at around 38% and 35%, respectively. Meanwhile, the remaining retailers had around 32% and 33% of its products in this price range.

Regarding the products costing 15-25 million VND, the share of products in CellphoneS, FPT Shop and The Gioi Di Dong was the highest at around 45% and 46%, while that of the remaining retailers accounted for around 32,5%.

In terms of products with an above 45 million VND price tag, their share was the highest in An Phat PC, Gear VN and Phong Vu at above 20%, while that of the remaining retailers was below 17%. The percentage of products in the price range of less than 15 million VND were mostly below 10%, with the exception of CellphoneS, accounting for 11,64%.

Overall, the general retail chains (The Gioi Di Dong, CellphoneS, FPT Shop) were highly accessible, capturing the mainstream and lower-income markets. In contrast, specialized IT and gaming stores (Gear VN, Phong Vu, An Phat PC) skewed significantly toward wealthier buyers looking for premium laptops.

Objective 3: An assessment of the comprehensive pricing strategies of retail chains.

The aim of this is to identify which retailers specialize in low-cost goods, which are positioned as premium, and which offer the widest price range (serving all demographics) without having to cross-reference every single laptop model.

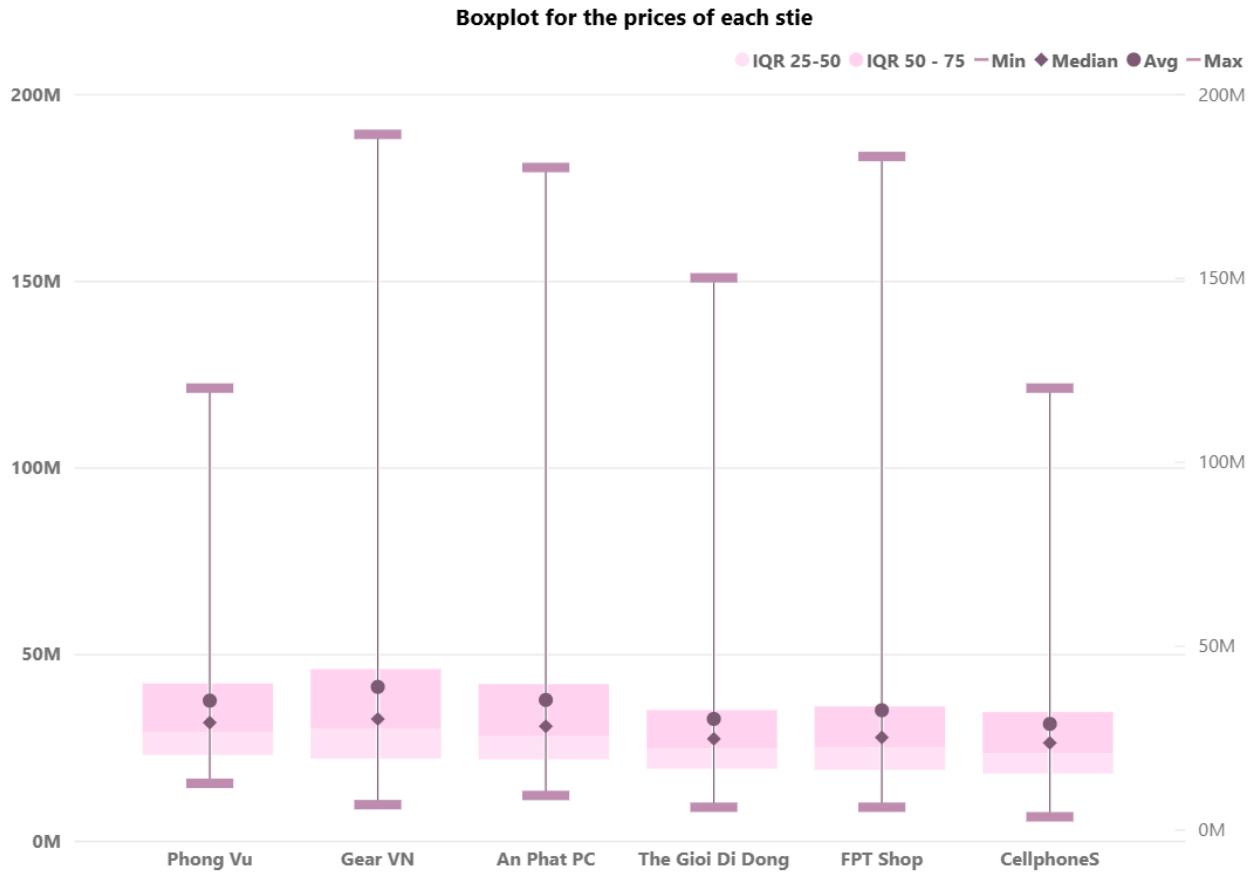

Figure 4: The price distribution of laptops based on their retailers

Regarding the Interquartile Range (IQR) and the median (represented by the diamond), Gear VN exhibited the highest median price at nearly 30 million VND. This was then followed by Phong Vu and An Phat PC at around 27 million VND. Furthermore, IQR elevation was the highest in these three retailers, indicating that these retailers offered products concentrated at a higher price tier compared to other competitors. Meanwhile, the median in the remaining retailers was lower than 25 million VND and their IQR positioning was slightly lower, suggesting that they offer products at a lower price tier, catering to the mass market.

Regarding the dispersion of each market, FPT Shop and Gear VN had the highest price dispersion with its most expensive products selling for nearly 190 million VND. This indicates that these markets serve a highly diverse demographic from budget-conscious to ultra-premium buyers. Meanwhile, Phong Vu and CellphoneS exhibited a lower price dispersion with its highest reaching around 125 million VND. This indicates that these

markets have a specialized inventory that avoids overly-priced products found in Gear VN and An Phat PC.

An interesting point to note is that the average price in each market was higher than their respective median, showing that each retailer carries a selection of extremely high-priced "outlier" products that pull the average upward.

Objective 4: Analysis of AI and non-AI laptops

The aim of this objective is to evaluate if there was a price gap between laptops with and without AI features

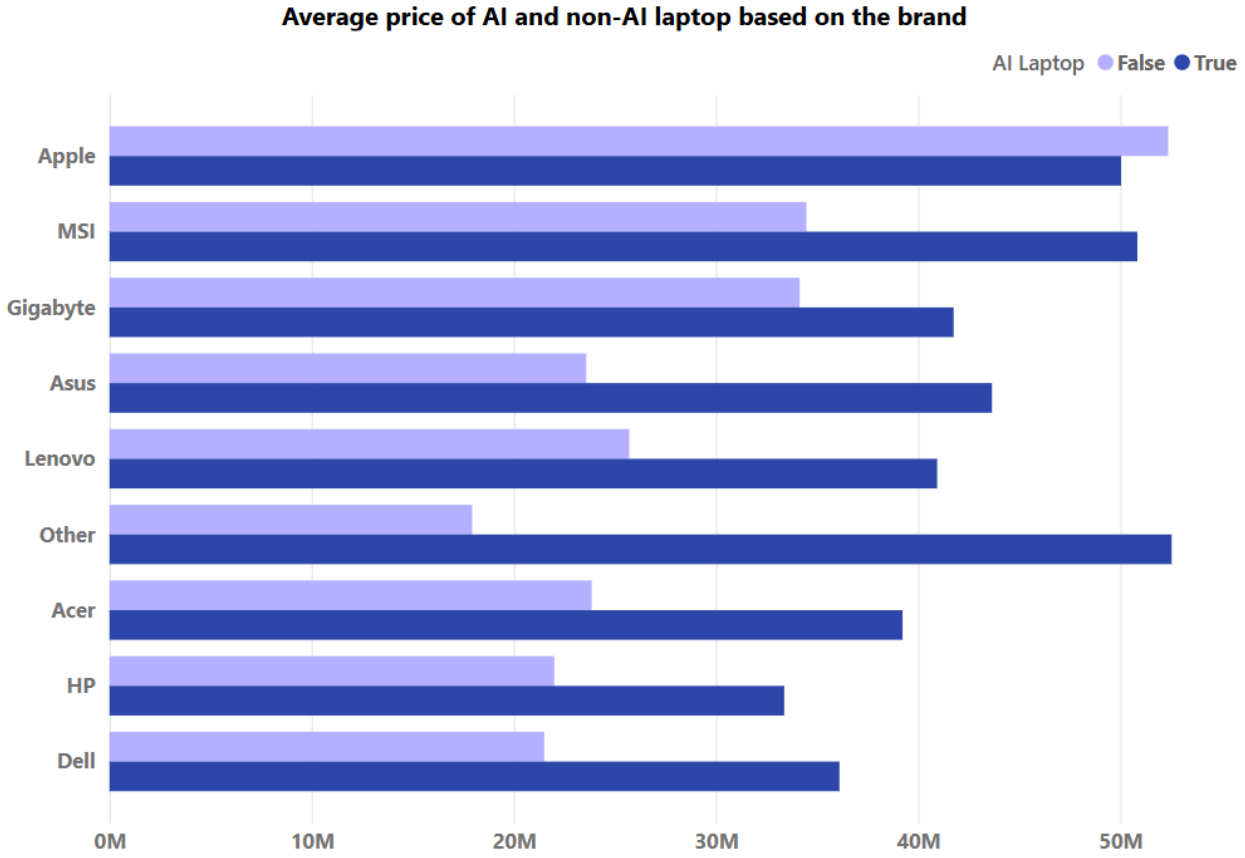

Figure 5: The price of AI and non-AI laptops for different brands on average

For almost every PC manufacturer listed, the bar representing laptops with AI features extended noticeably further to the right than that of non-AI laptops.

The price increase was substantial for many brands. Brands like Asus, Lenovo, and other smaller manufacturers had their non-AI laptops averaging in the 18-25 million VND price range, while their AI-equipped counterparts jumped drastically to the 40-50+ price range.

The only brand that differed from the overall market trend was Apple. According to the chart, Apple's non-AI laptops actually have a slightly higher average price (peaking just

above 52 million VND) compared to their AI-capable models which had their prices around 50 million VND.
Overall, with the single exception of Apple, the introduction of AI features is strongly correlated with a higher average price tag.

Objective 5: Analyze in the price distribution of products during the transition to RAM 32 GB

The aim of this objective is to help retailers to determine whether the price of laptop increase with more RAm

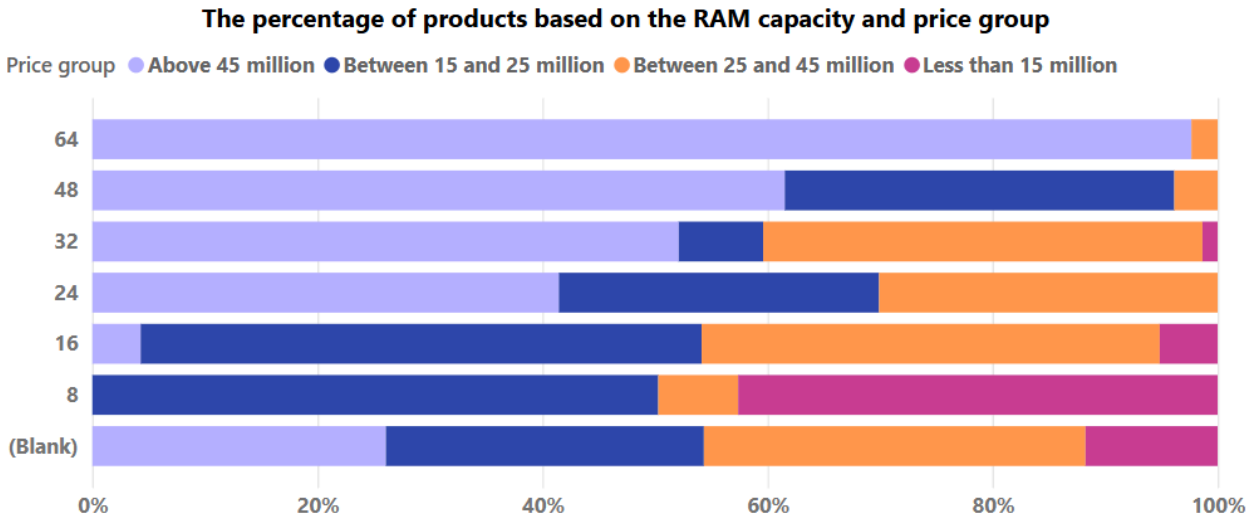

Figure 6: The distribution of products based on the RAM and price group

Regarding the 16GB laptops, the majority of its product was in the price range of 15-25 million at nearly half, which was greater than that of between 25 million VND and 45 million VND by 10%.
Meanwhile, the data for products above 45 million VND and below 15 million VND was around 5% each.
In terms of RAM 32 GB, the percentage of products in the above 45 million VND price range was the highest at 52,12%, which was followed by between 25 and 45 million VND, between 15 and 25 million and below 15 million VND at 39,04%, 7,5% and 1,35%.
Regarding laptops with RAM below 16 GB, the percentage of products in the less than 15 million VND category was the lowest. In terms of laptops above 32 GB, the majority of products fell in the above 45 million VND category.
Overall, there was a clear increase in the selling price of laptops the more RAM they have. Laptops with 8GB and 16GB RAM were on the lower end in terms of prices and

the more RAM they had the higher the prices were, reaching the peak at 64GB RAM laptops with a price tag of above 45 million VND dominating this RAM range.

Objective 6: Determine the strengths of each brand

The aim of this objective is to provide the details of each brand

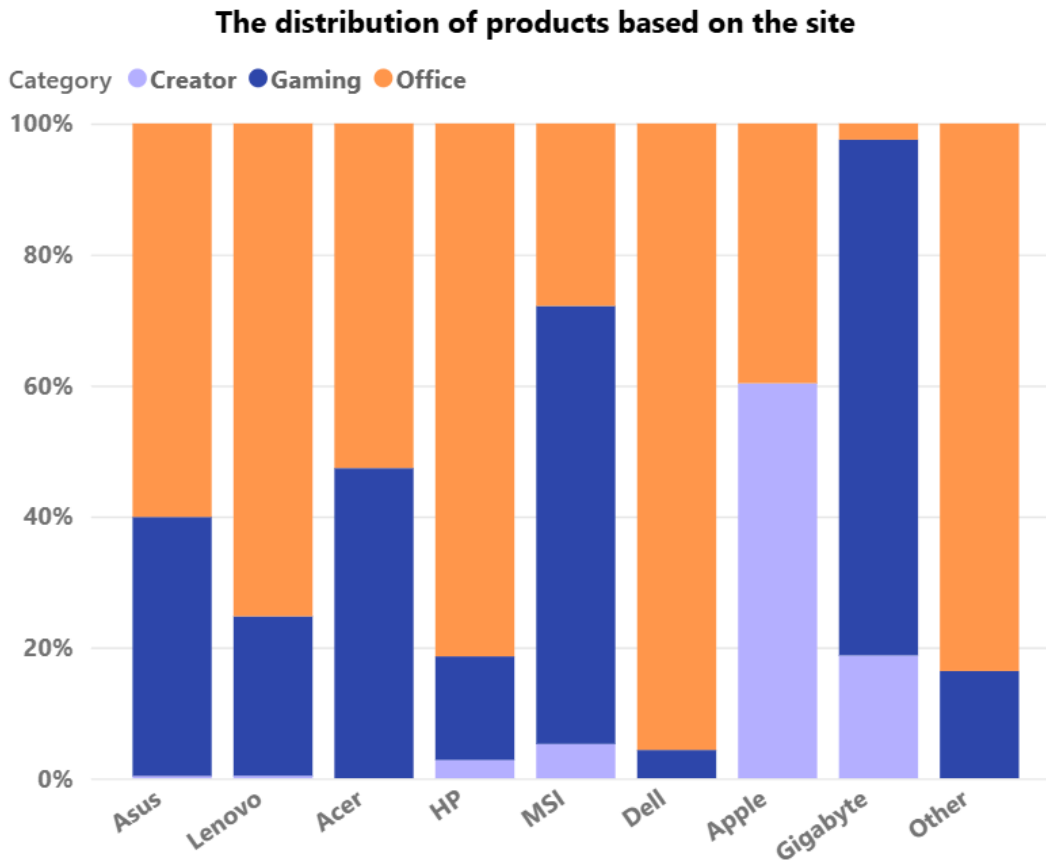

Figure 7: The distribution of products based on the site

Regarding ASUS, Lenovo, HP, Dell, it can be inferred that these four brands primarily focus on Office laptops due to its high share of these laptops produced and its large gap compared to other types of laptops. Acer, despite having a higher share of Office laptops, its gap was comparatively smaller (at 5%). Meanwhile, the share of MSI and Gigabyte Gaming laptops was significantly higher than the remaining categories. An interesting point to take note is that only four brands produced creator laptops (HP, MSI, Gigabyte and Apple) and Apple recorded the highest share of these laptops compared to the other type of laptop it produced.

Objective 7: The distribution of brand based on their prices and their types of RAM

The aim of this objective is to categorize the product into 'Casual', 'Middle-class' and 'High-end', which helps the retailer set up their product catalog suitable to their needs

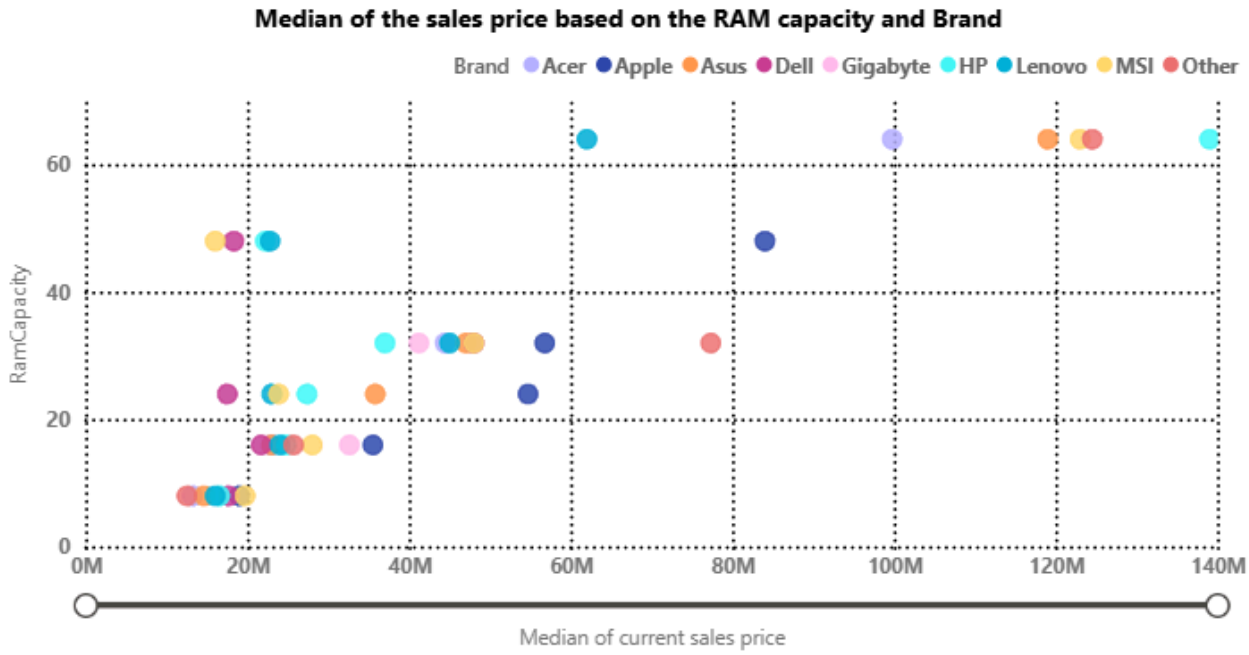

Figure 8: Median of the sales price based on the RAM capacity and Brand

In the scatterplot, it can be seen that laptops with RAM below or equal to 16 GB had a median of around 15 million to 35 million VND. Meanwhile, the median price of laptops with 24 GB ram is more sparse with Apple having the highest median at 55 million. Regarding those with 32GB RAM, there is one outlier having a median of nearly 80 million while the rest is scattered at around the 40 and 60 million VND range. Regarding 48GB RAM, the price for each brand had a median in the 20 million VND mark but Apple had a significantly higher median at nearly 90 million. Finally, the price dispersion for laptops with 64GB RAM was the highest with its lowest being at 60 million (Lenovo) while that of HP had a median of nearly 140 million.

Objective 8: Analyze the number of satisfied customers based on brand and type of laptops

The aim of this objective is to act as a metric to measure how satisfied consumers are with specific type of brand and type, allowing the retailer to know which product to stock up

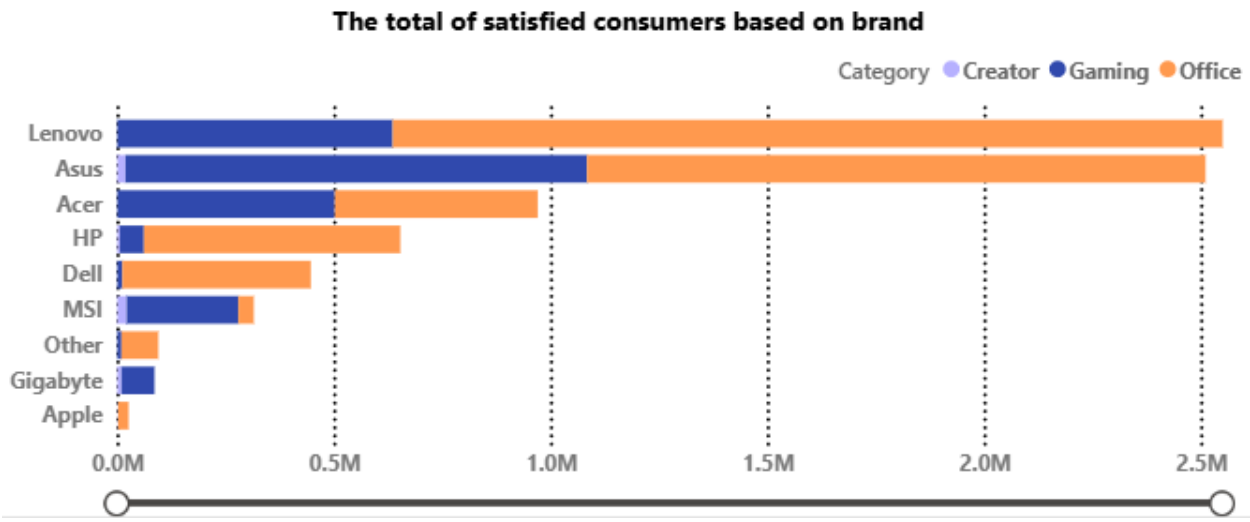

Figure 9.1: The total of satisfied consumers based on brand

Regarding Lenovo, Asus, HP and Dell, the number of satisfied customers for office laptops is significantly higher than that of gaming and creator laptops, with the number of satisfied customers for Lenovo Office laptops being the highest at nearly 1.8 million. Meanwhile, the number of satisfied customers for gaming laptops was the highest for MSI, Gigabyte and Acer. Another point worth mentioning is that Asus, MSI and HP had satisfied customers on their creator laptops but their figures are rather negligible compared to other types of office.

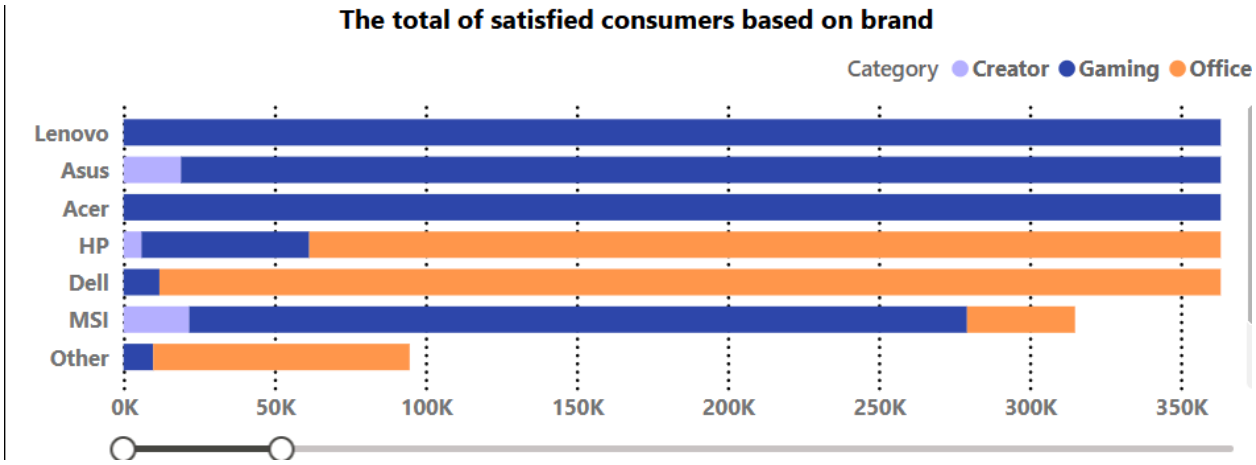

Figure 9.2: The total of satisfied consumers based on brand (smaller scale)

Objective 9: Analyze the need of RAM based on the market

The aim of this objective is to prove that if the trend of importing RAM 32GB laptop is suitable with the demand of consumers

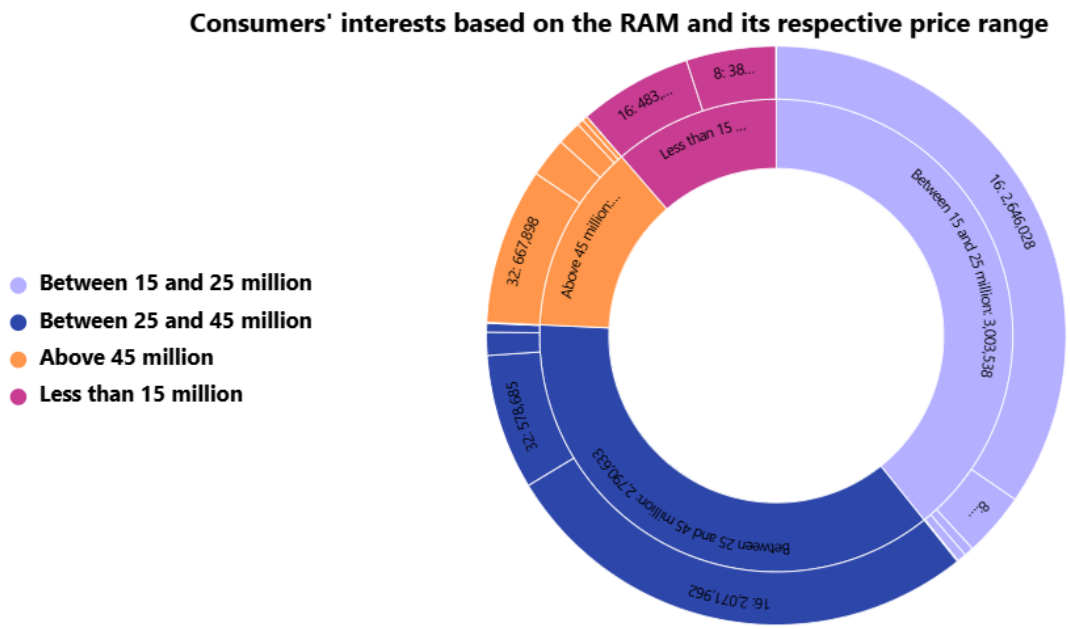

Figure 10: Customers' interests based on the RAM and its respective price range

In the price range of below 15 million VND, consumers are most interested in 16GB RAM laptops but the number of these consumers interested in this type of laptop in this price range was lower than that of products in the price range between 15 and 25 million and between 25 million and 45 million. Interestingly, in these price ranges, consumers still prefer those with 16GB RAM. Regarding laptops with 32GB RAM, consumers were generally interested in them if they were in the price range of above 45 million or between 25 and 45 million

Objective 10: Analyze the satisfactory rate (rating) based on the products in different price groups

The aim is to find out the price range that has the lowest rating, thereby determining the "market gap" so that retailers can stock up on models with higher quality

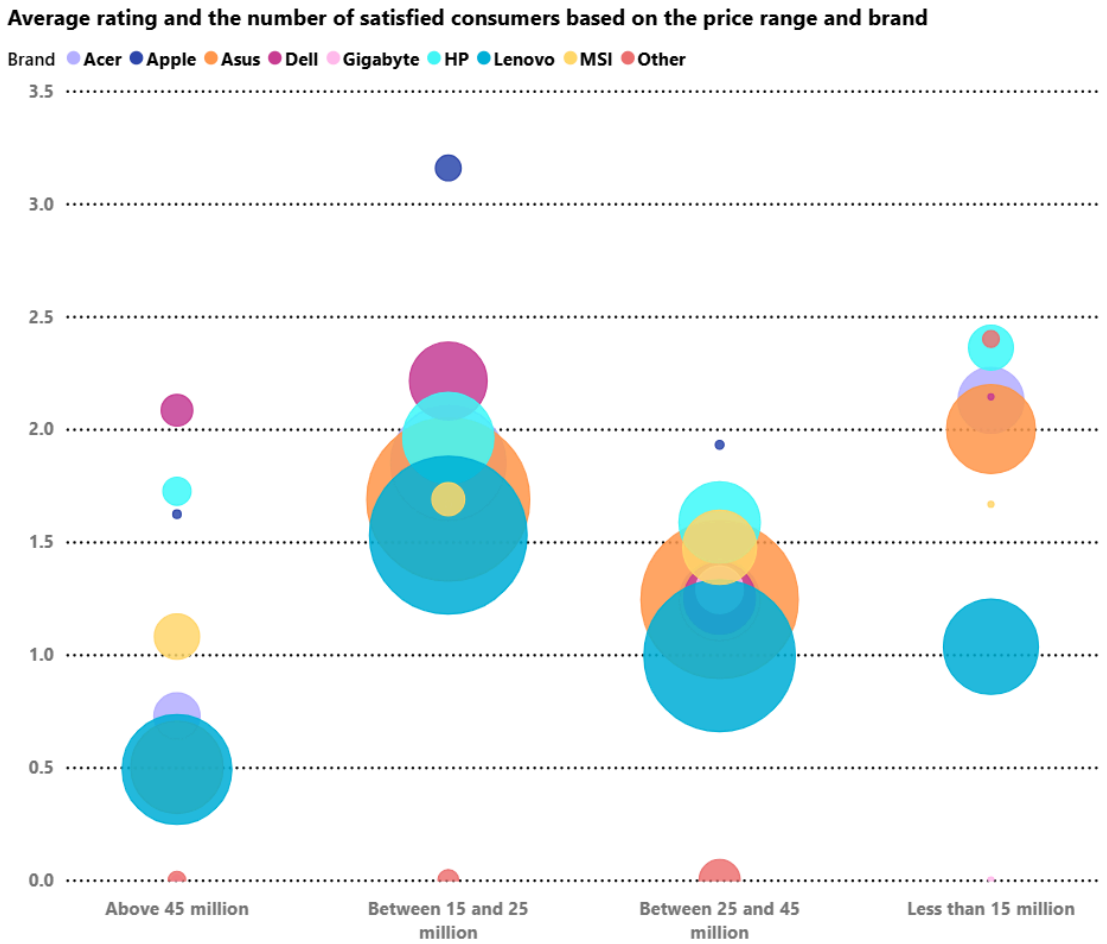

Figure 11: Average rating and the number of satisfied consumers based on price range and brand

The above 45 million VND price range represents the most critical market gap. Despite being the ultra-premium tier, it houses the lowest average ratings on the chart. Most notably, the largest volume of consumers in this bracket (the cyan Lenovo bubble) reports an exceptionally low average rating of around 0.5 out of 5.

The 25-45 million VND bracket also shows surprisingly poor performance. The bulk of its consumer volume (dominated by Lenovo andAsus) is clustered around a low 1.0 to 1.5 average rating.

Interestingly, customer satisfaction is significantly higher in the more affordable tiers. The less than 15 million VND and 15-25 million VND brackets show the majority of brands achieving average ratings between 1.5 and 2.5, with Apple peaking above 3.0 in the 15-25 million VND range.

Overall, customers paying top-tier prices (over 25 million, and especially over 45 million) are the least satisfied with their purchases.

Objective 11: Predicting the "Fair Market Price" and Identifying High-Value Bargains

The aim of this objective is to apply a Random Forest regression model to determine the "Fair Market Price" of laptops by analyzing their internal hardware components. By comparing these predicted values with actual store prices, the retailer can automatically generate an "Action List" of models to promote as "Best Deals"—specifically targeting units where the hardware value significantly exceeds the current retail price.

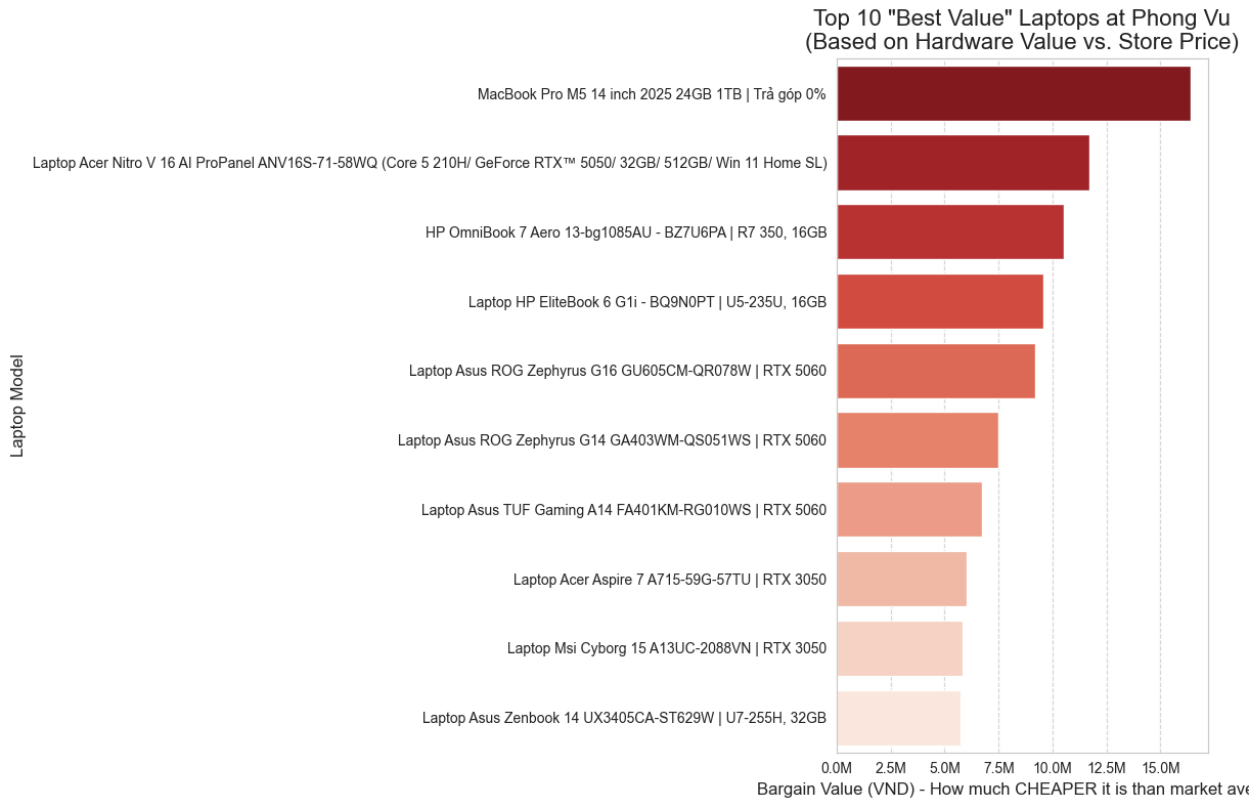

Figure 12: Top 10 "Best Value" Laptops at Phong Vu (Hardware Value vs. Store Price)

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--- PHONG VU ACTION LIST: MODELS TO PROMOTE AS 'BEST DEALS' ---
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	ProductName	CurrentPrice	Predicted_Price	Bargain_Value
2117	MacBook Pro M5 14 inch 2025 24GB 1TB | Trả góp 0%	50.0M	66.4M	16.4M
2195	Laptop Acer Nitro V 16 AI ProPanel ANV16S-71-5...	40.0M	51.7M	11.7M
2213	HP OmniBook 7 Aero 13-bg1085AU - BZ7U6PA | R7 ...	29.8M	40.3M	10.5M
2001	Laptop HP EliteBook 6 G1i - BQ9N0PT | U5-235U,...	32.4M	42.0M	9.6M
2002	Laptop Asus ROG Zephyrus G16 GU605CM-QR078W | ...	62.0M	71.2M	9.2M

The MacBook Pro M5 14-inch (2025) is identified as the highest priority for promotion, with a current price of 50.0 million VND against a predicted hardware value of 66.4 million VND, representing a massive **16.4 million VND** bargain. Following this, the Acer Nitro V 16 AI ProPanel shows a significant value gap, where its predicted worth of 51.7 million VND is much higher than its 40.0 million VND price tag, offering a **11.7 million VND** saving. In the ultra-portable and business categories, the HP OmniBook 7 Aero and HP EliteBook 6 G1i both exhibit strong bargain values of **10.5 million** and **9.6 million VND** respectively, as their retail prices (29.8M and 32.4M) sit well below their calculated market worth. Finally, the Asus ROG Zephyrus G16, despite being a premium gaming model priced at 62.0 million VND, is valued by the model at 71.2 million VND, leaving a **9.2 million VND** margin. These findings indicate that the retailer can leverage these specific models in "Best Deal" marketing campaigns, as they represent the most significant hardware-to-price advantages currently available in the inventory.

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